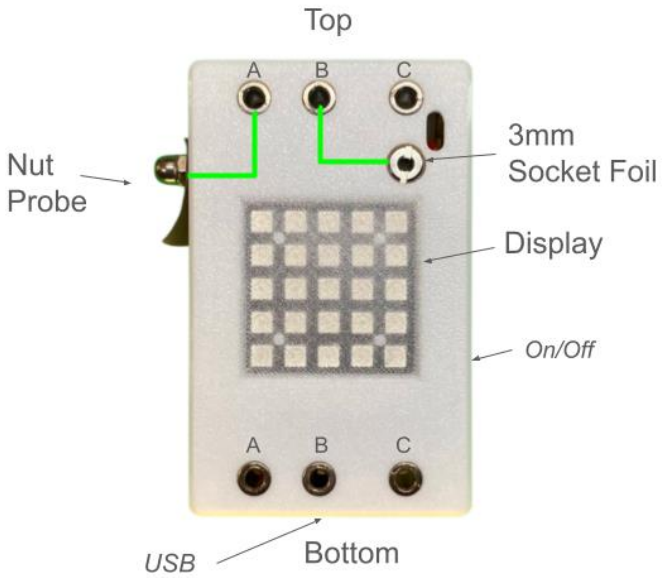
Using the Tester

Device Description & Charging

Form factor: Rectangular device with two connections (“bottom” and “top”). From version 3.0 a PRO version with a small OLED screen in which exact values of the resistances are shown. For the rest the functionality of both versions is the same.





STANDARD **PRO**

Bottom Face (to the part of the body cord connected to cable reel)

- Three 4 mm sockets labelled A-B-C (left to right): B centered; C is 20 mm to the right of B; A is 15 mm to the left of B.
- Used to connect the body wire plug that connects to the reel.

Top Face (to the part of the body cord connected to the weapon)

- Four sockets in total: three 4 mm sockets labelled A–B–C (as above) and one 3 mm socket to connect the epee or foil/sabre body wire plug that connects to the weapon.

Controls & Fittings

- Power button (on/off).
- M5 cap nut on the left side (when the bottom connection is towards the user).

Power Supply

- Rechargeable 3.7 V Li-ion battery.
- Recharge via the USB-C port on the front of the bottom face when the battery is depleted. A blue led will light when the battery is fully loaded.

Status Display

A **5×5 LED matrix** between the two faces indicates operating modes and diagnostics using four colours:

- Green — the item under test works and complies with FIE regulations.
- Yellow — the item works, and it is likely that it will be accepted through weapon control, but values are too close to the limit, therefore we recommend the user to repair it.
- Orange — the item works but does not comply with FIE regulations.
- Red — the item doesn't work.

The Pro version has an additional OLED display showing the exact values of the resistance.

Operating Principle (Button-less Switching)

The tester is operated without any buttons other than the power button. A base mode—called the Body Cord Tester mode (a pass-through/continuity test)—is used to initialise measurements. Modes are changed by making specific connection combinations between sockets. To switch between any two modes, you must first return to Body Cord Tester mode, except in the call room mode.

Modes of Operation

Body Cord Tester Mode

- On power-up the device starts in Body Cord Tester mode. A yellow/orange “snake” animation runs during initialisation; afterwards a blinking dot is shown. Green blinking dot indicates that the device is calibrated, a red blinking dot indicates that the device is operating with default calibration, consult the building manual to proceed with the calibration.
- Solid “green lines” indicate a working connection with resistance < 1 Ω (FIE-compliant).
- Solid yellow lines indicate a working connection with resistance between 1 and 3 Ω (close to FIE regulations).
- Solid orange lines indicate a working connection with resistance between 3 and 10Ω (not FIE-compliant).
- Broken red lines indicate resistance > 10 Ω (connection likely open; body cord may be faulty).

When a body cord passes the test (all three connections “green”), the blinking dot turns blue. The measured resistances are stored and subtracted from subsequent measurements in other modes.

Measurement proceeds in two steps:

Step 1: After the body wire to be tested is plugged in at both sides, the resistance of every single wire is accurately measured. Short circuits or wrong connections between top and bottom connector are indicated with blue lines indicating the problem.

Step 2: After step 1 the device switches to fast mode until all cables are disconnected to allow the detection of “micro breaks”. The user may move and stress the connections to detect problems.

Once the body wire is found to be good, it can be used to perform the other tests (weapons & lamé).

Épée Mode

Test the body wire first. Connect the épée to a body cord plugged into the bottom connection. The device enters Épée Mode when the tip is pressed (assuming both body cord and weapon are functional and its resistance lower than 20 Ohm). An “E” is displayed on the LED matrix (white when the tip is not pressed). Measure circuit resistance by pressing the tip against a non-conductive surface (épée circuit only) or against the M5 cap nut (path from épée to tip).

- Green “E” — resistance < 2 Ω (FIE-compliant).
- Yellow “E” — resistance between 2 and 4 Ω (close to FIE regulations).
- Orange “E” — resistance > 4 Ω (not FIE-compliant).
- White “E” — resistance > 20 Ω (épée not working).
- Blue lines — short-circuit detected.

To return to Body Cord Tester mode, short-circuit any bottom socket with socket B or C on the top face.

Foil Mode

Connect the foil to a body cord plugged into the bottom connection. The tester enters Foil Mode automatically (if its resistance is lower than 20 Ohm). A green “F” is displayed; pressing the tip against a non-conductive surface turns the “F” into a white square (circuit debounce time: 10 ms).

- Green “F” — resistance < 2 Ω (FIE-compliant).
- Yellow “F” — resistance between 2 and 4 Ω (close to FIE regulations).
- Orange “F” — resistance > 4 Ω (not FIE-compliant).
- Blue lines — short-circuit detected.

To measure the full path (socket → foil → tip), press the tip against the M5 cap nut:

- Green square — resistance < 1 Ω (FIE-compliant).
- Yellow square — resistance between 1 and 2 Ω (close to FIE regulations).
- Orange square — resistance > 2 Ω (not FIE-compliant).

Pressing the tip against a non-conductive surface shows a white light.

To return to Body Cord Tester mode, short-circuit any bottom socket with socket B or C on the top face.

Expert Foil Tip Leak Test

If Expert Leak Test Mode is enabled in the Advanced Setup (see below), the tester checks for insulation leakage when the tip is pressed. In this mode, a blue square is displayed if the measured resistance is below 500 Ω, indicating a leakage path.

Connect the foil to the tester press the tip and touch the M5 cap nut with the body cord crocodile clip. The tester enters Leak Test Mode. To check the insulation between the tip and the barrel, wrap a small piece of lamé around the tip and press the tip down.

- Blue square ON — insulation leakage detected (tip/barrel insulation faulty).
- Green arrow pointing bottom — insulation is satisfactory.

Keep the foil tip depressed and touch the M5 cap nut with the crocodile clip again. The tester returns to Standby Mode.

Lamé Testing Mode

Connect the lamé to the crocodile clip of a foil body cord connected to the bottom connection, then touch it either with the 4 mm pin of the body cord or with the M5 cap nut. The device enters Lamé Testing Mode, displaying a diamond on the LED matrix.

- Green diamond — lamé in contact with the chosen connection and resistance < 5 Ω (FIE-compliant).
- Yellow diamond — lamé in contact with the chosen connection and resistance between 5 and 10 Ω (close to FIE regulations).
- Orange diamond — lamé in contact with the chosen connection and resistance between 10 and 20 Ω (not FIE-compliant).
- Red diamond — resistance > 20 Ω (lamé not working).

To return to Body Cord Tester mode, short-circuit any bottom socket with socket A or B on the top face.

Guard Testing Mode

Connect a weapon to the bottom connection and touch the guard with the M5 cap nut. The device enters Guard Testing Mode, displaying a cross on the LED matrix.

- Green cross — resistance < 5 Ω (FIE-compliant).
- Yellow cross — resistance between 5 and 8 Ω (close to FIE regulations).
- Orange cross — resistance between 8 and 20 Ω (not FIE-compliant).
- Red cross — resistance > 20 Ω (guard not working).

To return to Body Cord Tester mode, short-circuit any bottom socket with socket A or B on the top face.

Call room Foil/Sabre Testing Mode

If both the weapon and the lamé of a foil or sabre fencer need to be tested in the call room, the tester allows you to switch easily between Foil/Sabre Mode and Lamé Mode without changing the connections.

Connect the bottom plug of the body cord to the tester. The tester automatically enters Foil/Sabre Mode, allowing the resistance of the weapon circuit to be checked.

Once the weapon has been tested, touch the lamé with the M5 cap nut for approximately 2 seconds. The tester automatically switches to Lamé Mode, allowing the resistance of the lamé to be measured.

To return to Foil/Sabre Mode, simply touch the guard of the weapon with the M5 cap nut for approximately 2 seconds.

Cable Reel Testing Mode

Short-circuit sockets A and B on the top face. The device shows an “R” and enters Cable Reel Testing Mode. Connect the cable reel using two épée body cords; the tester behaves like Body Cord Tester mode with different thresholds.

- Solid green lines — resistance < 10 Ω (FIE-compliant).
- Solid yellow lines — resistance between 10 and 20 Ω (close to FIE regulations).
- Solid orange lines — resistance > 20 Ω (not FIE-compliant).
- Broken red lines — resistance > 50 Ω (connection likely open).

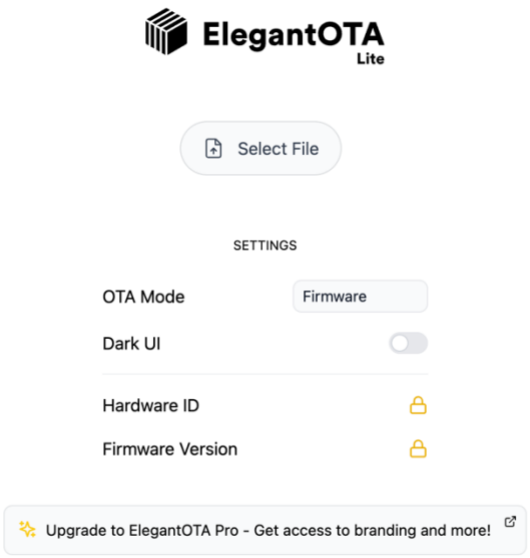
To return to Body Cord Tester mode, short-circuit sockets A and B on the top face.

Software updating.

The tester firmware can be updated over the air (OTA) via Wi-Fi. When the tester is switched on, it creates a Wi-Fi network named Tester that remains available for 90 seconds. Connect to this network using the default password:

Password: 01041967

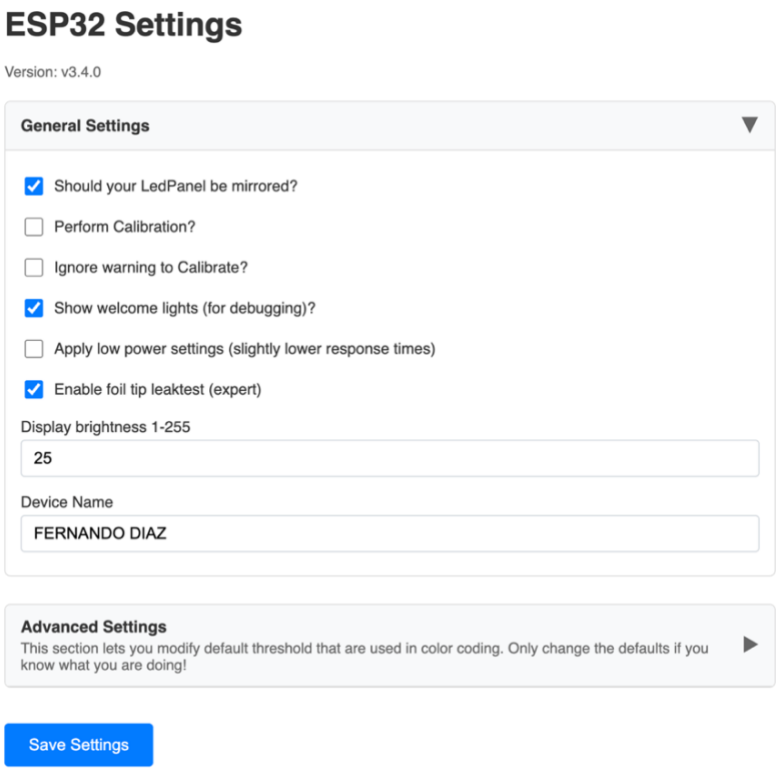
Open a web browser and navigate to OTA Update Portal: http://192.168.4.1/update



Select the latest OTA firmware file and upload it to update the tester. Do not switch off the device during the update process.

Advanced setup.

While connected to the tester's Wi-Fi network, open a web browser and navigate to Setup Portal: http://192.168.4.1/settings



The following configuration options are available:

- LED panel mirrored? Some LED panels are manufactured with the display mirrored. Enable this option if the displayed patterns appear reversed.
- Perform calibration: Forces the tester to perform a full calibration. Use this option only if you are familiar with the calibration procedure.
- Ignore calibration warning: Disables the calibration reminder displayed when the tester is operating with the default calibration values.
- Show welcome animation: Enables or disables the start-up "snake" animation. This option is mainly intended for debugging.
- Low-power mode: Reduces power consumption at the expense of slightly slower response times.
- Enable foil tip leak test (Expert): Enables the insulation leakage test described in Foil mode.

The Advanced Settings menu allows the resistance thresholds associated with each display colour to be modified.

Warning: These parameters determine the tester's measurement criteria. They should only be changed by users who fully understand their effect on the instrument's operation. Incorrect settings may lead to inaccurate test results.